

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10: SUB-STRAND 1.3 QUADRATIC EQUATIONS

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation for the sub-strand. Driving question: How can writing and solving a quadratic equation help us find unknown dimensions, quantities, and arrangements in real-life situations across Kenya? Think back to Achieng's chair problem at the Kisumu prize-giving ceremony and to your work in Lessons 1 to 8. Answer every question in the space under it. Show each step of your working, write your reasoning in full sentences, and give your final answers with units or a meaning where the question asks for one. Use r for the number of rows in Achieng's problem.

Part 1: Turning Achieng's seating problem into an equation

A school in Kisumu needs to arrange 120 chairs for guests in a rectangular block. The number of columns must exceed the number of rows by 7. (a) Write a Let statement for the number of rows and write an expression for the number of columns. (b) Write an equation that shows the total number of chairs. (c) Expand and rearrange it into the standard form $ax^2 + bx + c = 0$ and state the values of a, b and c. (d) Explain why Achieng could not solve this by simple division or guess-and-check, and why a squared term appears.

Write your answer here:

Part 2: Solving by factorisation

Use the equation you formed, $r^2 + 7r - 120 = 0$. (a) Use the product-and-sum method: state the product and the sum you need, and name the two numbers that work. (b) Write the equation in factorised form. (c) Use the Zero Product Property to find both values of r . (d) Check your factorisation by expanding the brackets.

Write your answer here:

Part 3: Verifying and choosing the valid answer

(a) Substitute each of your two roots into the original equation $r^2 + 7r - 120 = 0$ and show what you get. **(b)** Which root can Achieng use for the chair arrangement? Give a reason based on the situation. **(c)** State the number of rows and the number of columns, and show that they give the correct total number of chairs.

Write your answer here:

Part 4: Using the method in a new Kenyan context

A market trader in Kisumu has a rectangular display table with an area of 28 m^2 . The length of the table is 3 m more than its width. (a) Let w be the width in metres. Form a quadratic equation in standard form. (b) Solve it by factorisation and find both roots. (c) Decide which root is valid, using the context, and state the width and length of the table. (d) Show how you would verify your answer.

Write your answer here:

Part 5: Answering the driving question

Using Achieng's chair problem and at least one other Kenyan situation from the unit, explain how writing and solving a quadratic equation helps us find unknown dimensions, quantities or arrangements. Describe the full pathway from the real-life situation to the final answer, and explain why you must always check your answers in context.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Forming quadratic equations	Writes correct Let statements and expressions, forms the product equation, and rearranges it correctly into standard form with a, b and c identified, in every context.	Forms the equations correctly with minor slips in rearranging or in identifying a, b and c.	Has difficulty writing the expression for the second unknown or forms a linear equation or an equation not set to zero.
Solving by factorisation	Uses the product-and-sum method accurately, writes correct factors, applies the Zero Product Property to get both roots, and checks by expanding.	Factorises and finds both roots correctly but does not always check by expansion or explain the Zero Product Property.	Chooses the wrong factor pair, makes sign errors, or finds only one root.
Verification and contextual validity	Substitutes both roots, correctly rejects the invalid root with a clear reason from the context, and checks that the final dimensions give the known total.	Substitutes the roots and rejects the invalid root, but the reason or the final check is incomplete.	Does not verify by substitution or does not use the context to choose the valid answer.

Explaining the full journey and the driving question	Gives a clear, ordered pathway from real-life situation to verified answer, explains why the squared term appears, and links to at least two Kenyan contexts.	Describes most steps of the pathway with a Kenyan example but with limited explanation of why the squared term appears.	Lists some steps with little reasoning and few or no links to real-life situations.
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ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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